LECTURE 1

Course Overview

An overview of data science, Data200, and the data science lifecycle.

Presented by Dr. Devan Becker, Data200

Adapted from Data 100, Summer 2023 @ UC Berkeley by Bella Crouch and Dominic Liu

Content credit: Lisa Yan, Josh Hug, Suraj Rampure, Allen Shen, Joey Gonzalez, and Sam Lau

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Roadmap

- **Intros**
- What is data science?
- What will you learn in this class?
- Course overview
- Data Science Lifecycle
- Demo

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Instructors

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What is Data Science?

- Intros
- **What is data science?**
- What will you learn in this class?
- Course overview
- Data Science Lifecycle
- Demo

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Why Data Science Matters

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Data science enhances critical thinking

The world is complicated! Decisions are hard.

Data is used everywhere to answer hard questions and make tough decisions:

- Science
- Medicine
- Social science
- Engineering
- Sports

Claims about data come up in discussing almost any important issue:

- It is usually not easy to tell what the data “says”
- **Empower yourself** to participate in the arguments that shape your life and your society

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Technology Trends

- 2020s ?
- 2010s Data Industry
  - Collect and sell information
- 2000s Internet Industry
  - Online retailers and services
- 1990s Software Industry
  - Sold computer software
- 1980s Hardware Industry
  - Sold computers

From Joey Gonzalez.

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The darker side of data science?

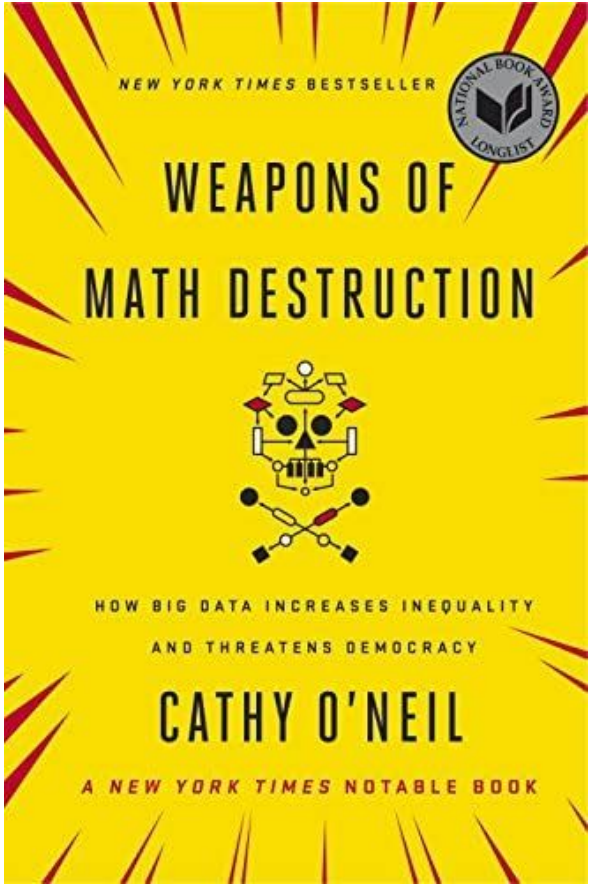
Obscuring complex decisions:

- Mortgage-backed securities → market crash
- Teaching scores & job advancement

Reinforcing historical trends and biases:

- Hiring based on previous hiring data
- Recidivism and racially biased sentencing
- Social media, news, and politics

We will discuss the ethics of data science throughout the class!



NPR author interview
with Cathy O'Neil

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But…we are optimistic!

Knowledge is empowering.

Data science offers **immense potential** to address challenging problems facing society.

The future is in your hands, and I believe:

You will use your knowledge for good.

I am thrilled to teach Data 200 :-)

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Data science enhances critical thinking

The world is complicated! Decisions are hard.

Data science is a fundamentally human-centered field that facilitates decision-making by quantitatively balancing tradeoffs.

- To quantify things **reliably** we must:
  - Find relevant data;
  - Recognize its limitations;
  - Ask the right questions;
  - Make reasonable assumptions;
  - Conduct an appropriate analysis; and
  - Synthesize and explain our insights.
- Apply **critical thinking and skepticism** at every step; and
- Consider how our decisions **affect others**.

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Our primary goal for you in this course

The world is complicated! Decisions are hard.

Data science is a fundamentally human-centered field that facilitates decision-making by quantitatively balancing tradeoffs.

- To quantify things **reliably** we must:
  - Find relevant data;
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  - Ask the right questions;
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  - Conduct an appropriate analysis; and
  - Synthesize and explain our insights.
- Apply **critical thinking and skepticism** at every step; and
- Consider how our decisions **affect others**.

After this course, you should be able to take data and produce useful insights on the world's most challenging and ambiguous problems.

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What is Data200?

PRINCIPLES AND TECHNIQUES OF DATA SCIENCE

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Data science is a fundamentally interdisciplinary field

Joey Gonzalez
(co-creator of this course)

Data Science is the application of data centric, computational, and inferential thinking to:

- Understand the world (science).
- Solve problems (engineering).

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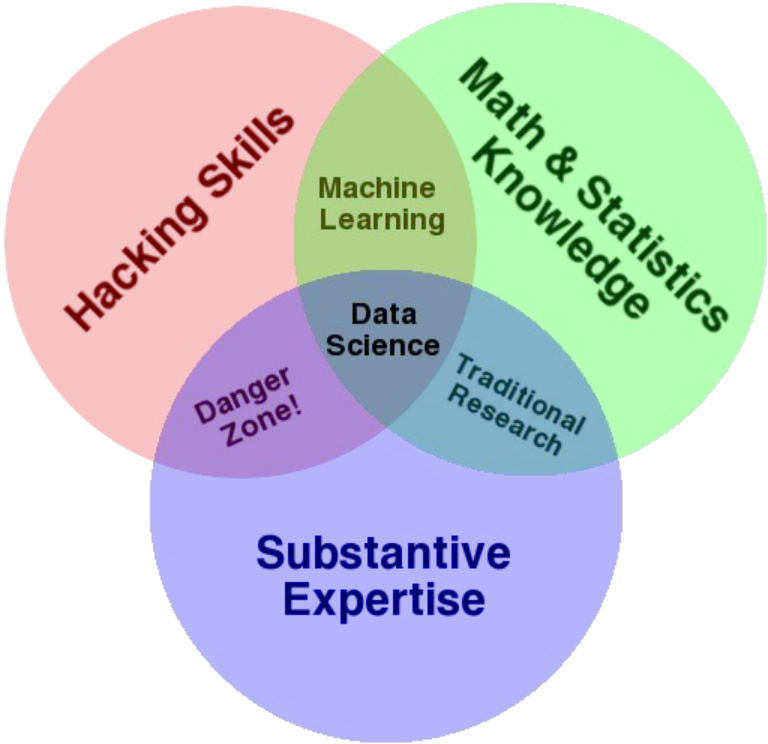
Data Science Venn Diagram



by Drew Conway in 2010 (link)

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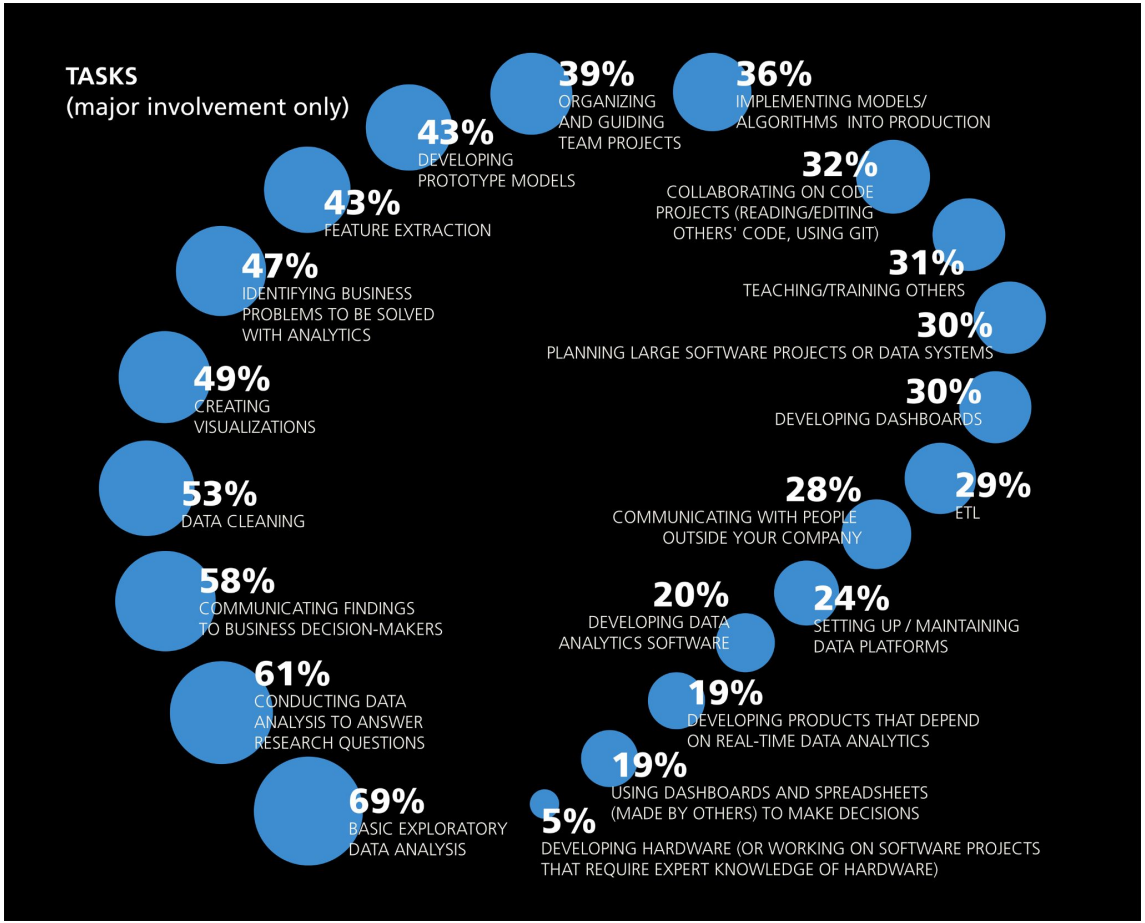
Data science in industry



The major tasks that data scientists say they work on regularly.

Self-reported. Based on the results of the [2016 Data Science Salary Survey](#).

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Data science requires engineering and scientific insight

Good data analysis is not:

- Simple application of a statistics recipe.
- Simple application of statistical software.

There are many **tools** out there for data science, but they are merely tools.

- **They don't do any of the important thinking!**

"The purpose of computing is insight, not numbers."

R. Hamming. *Numerical Methods for Scientists and Engineers (1962).*

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Example Questions in Data Science

Some (broad) questions we might try to answer with data science:

- What show should we recommend to our user to watch?
- In which markets should we focus our advertising campaign?
- Is the use of the COMPAS algorithm for prison sentencing fair?
- Should I send my kids to daycare?
- Is the world getting better or worse?
- What areas of the world are at higher risks for climate change impact in 10 years? 20?
- Where should we put docking ports for our bikes?
- What should we eat to avoid dying early of heart disease?
- Do immigrants from poor countries have a positive or negative impact on the economy?

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What will you learn in this class?

- Intros
- What is data science?
- **What will you learn in this class?**
- Course overview
  - Lots of important details
- Data Science Lifecycle
- Demo

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What is Data200?

PRINCIPLES AND TECHNIQUES OF DATA SCIENCE

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Course goals

Prepare

Prepare students for advanced Berkeley courses in **data management**, **machine learning**, and **statistics**, by providing the necessary foundation and context.

Enable

Enable students to start careers as data scientists by providing experience working with **real-world data, tools, and techniques**.

Empower

Empower students to apply computational and inferential thinking to address **real-world problems**.

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Tentative List of Topics to be Covered in Data200

- Pandas and NumPy
- Relational Databases & SQL
- Exploratory Data Analysis
- Regular Expressions
- Visualization
  - matplotlib
  - Seaborn
  - plotly
- Sampling
- Probability and random variables
- Model design and loss formulation
- Linear Regression
- Feature Engineering
- Regularization, Bias-Variance Tradeoff, Cross-Validation
- Gradient Descent
- Logistic Regression
- Clustering
- PCA

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Prerequisites

Official prerequisites for this course:

- Prerequisite: Data 100.
- Co-requisites: one of ST230, ST231, ST260, or equivalent

The prereqs are being strictly enforced! We will **not** be teaching:

- How to use Python.
- How to use Jupyter notebooks.
- Inference topics from ST*.
- Linear algebra (though we will review this topic to a greater degree).

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Course Components

- **Assignments x4**
- **Labs x5**
- **Participation**
  - **Class**
- **Midterm Project**
- **Final Project**
- **Final Exam**

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Assignments

- Designed to get you working with *real data*.

- Handed out as Jupyter Notebooks
  - Use wlu.syzygy.ca to avoid installing things
  - *OPTIONAL*: Use Positron, VSCode, or Jupyter on your own device.

- Submitted to GradeScope as i[ynb

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Participation

- **Quiz**
  - Quiz questions shown in slides
  - Answers on iClicker

- **Autograded**
  - .ipynb distributed before class
  - Hand in within 48 hours

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Midterm and Final Projects

Show me what you know!!!

- Select a topic that is meaningful to you, then apply the course materials!
- Groups of 2-5 people.
  - No groups of 1, no groups of 6+
- Well-formatted ipynb notebooks!
  - There are marks taken off for poor formatting!

Midterm Project

- Data cleaning (preferably including pivot tables), regex, and visualization.

Final Project

- Modelling and interpretation, especially regarding bias-variance tradeoff.
- Can be a continuation of midterm (doesn't have to be).

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Online platforms

Course website (MyLS)

- Where all lectures, assignments, and participation notebooks are posted.

Syzygy (wlu.syzygy.ca)

- Where you will work on all assignments and projects
  - You will have to upload the assignments yourself.

Gradescope (gradescope.com)

- Where all assignments are submitted.

Lecture Notes (https://ds100.org/course-notes/)

- A summary of the highlights of topics covered in each lecture.

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Programming Environment for our Course: wlu.Syzygy.ca

File Edit View Run Kernel Tabs Settings Help

Files | Running | Commands | Cell Tools | Tabs

> notebooks > transit-zurich

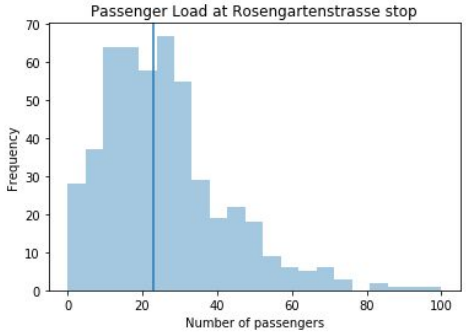
Name	Last Modified
transit.ipynb	2 minutes ago
passenger.csv	2 hours ago
routes.json	2 hours ago
stops.json	2 hours ago

transit.ipynb

Code | Python 3

We plot the number of passengers at the Rosengartenstrasse stop.

In [93]:
```
load = df[df.stopNameShort=='ROSE'].passengerLoadStop
sns.distplot(load, kde=False)
plt.axvline(load.median())
plt.title('Passenger Load at Rosengartenstrasse stop')
plt.xlabel('Number of passengers');plt.ylabel('Frequency');
```



Compare the median load at this stop with the medians of all stops.

In [94]:
```
sns.distplot(df.groupby('stopNameShort')
             .passengerLoadStop.median(), kde=False)
plt.axvline(load.median())
plt.title('Passenger load medians across all stops');
plt.xlabel('Median passenger load')
plt.ylabel('Frequency');
```

Passenger load medians across all stops

120

routes.json

stops.json | routes.json

```
▾ 564: {} 3 keys
    type: "Feature"
  ▾ properties: {} 4 keys
      stopId: 2749
      stopNumber: 2104
      stopNameShort: "ROSE"
      stopName: "Zürich, Rosengartenstrasse"
  ▸ geometry: {} 2 keys
```

passenger.csv

Delimiter: ,

stopSequer	stopId	stopNameShort	stopName
5	2104	ROSE	Zürich, Rosengartenstra
6	564	BUCH	Zürich, Bucheggplatz
7	2017	RADI	Zürich, Radiostudio
8	498	BIRD	Zürich, Birchdörfli
9	1705	NEUA	Zürich, Neuaffoltern
10	1000	GLAU	Zürich, Glaubtenstrasse
11	767	EINF	Zürich, Einfangstrasse

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Data Science Lifecycle

- Intros
- What is data science?
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- Demo

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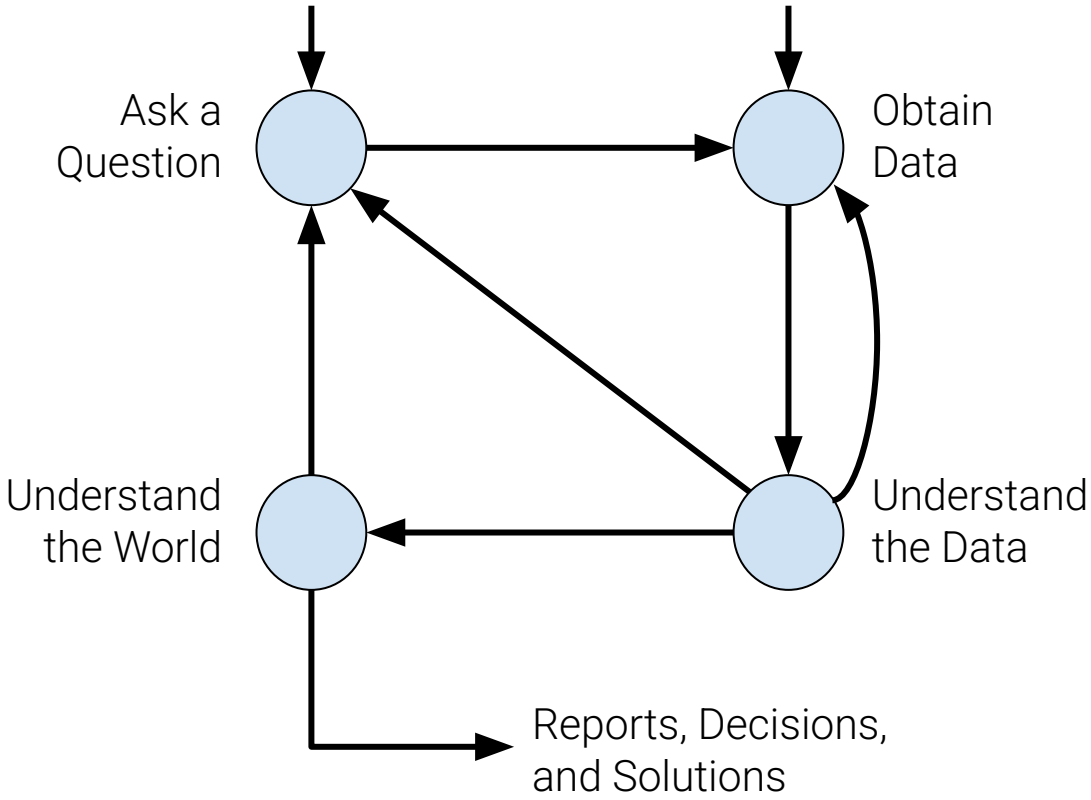
Data science lifecycle

The data science lifecycle is a **high-level description** of the data science workflow.

Note the two distinct entry points!



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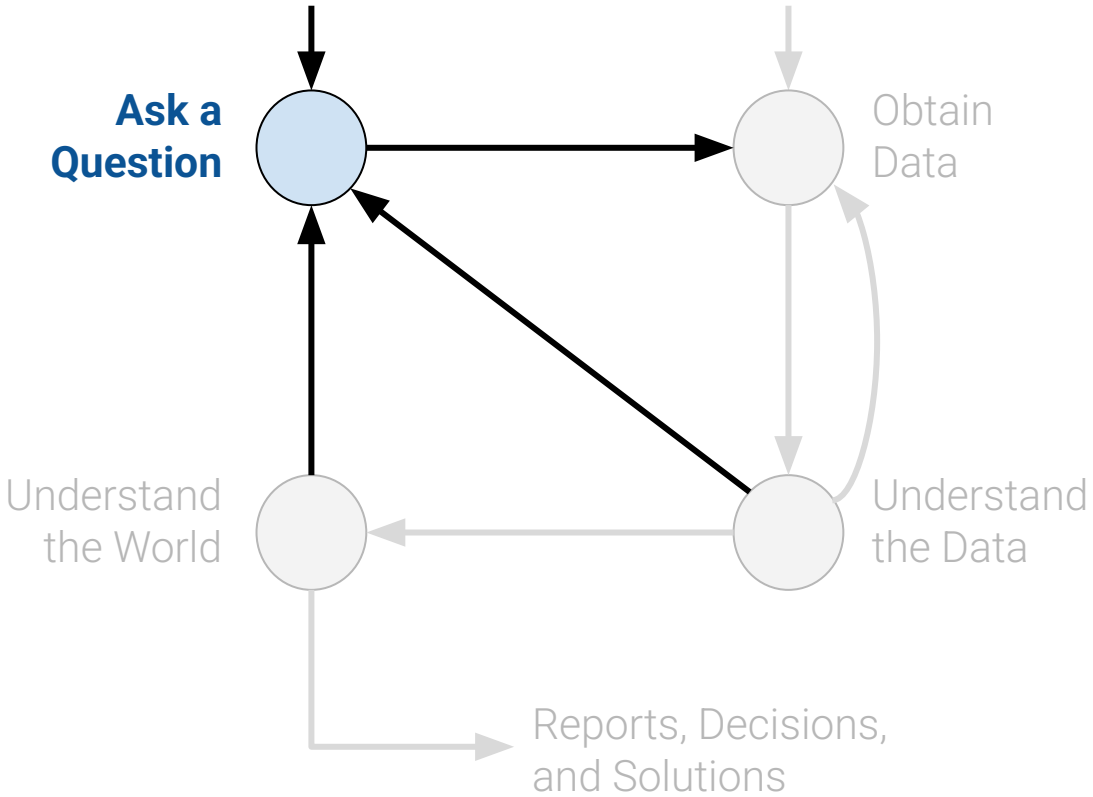
1. Question/Problem Formulation

- What do we want to know?
- What problems are we trying to solve?
- What hypotheses do we want to test?
- What are our metrics for success?



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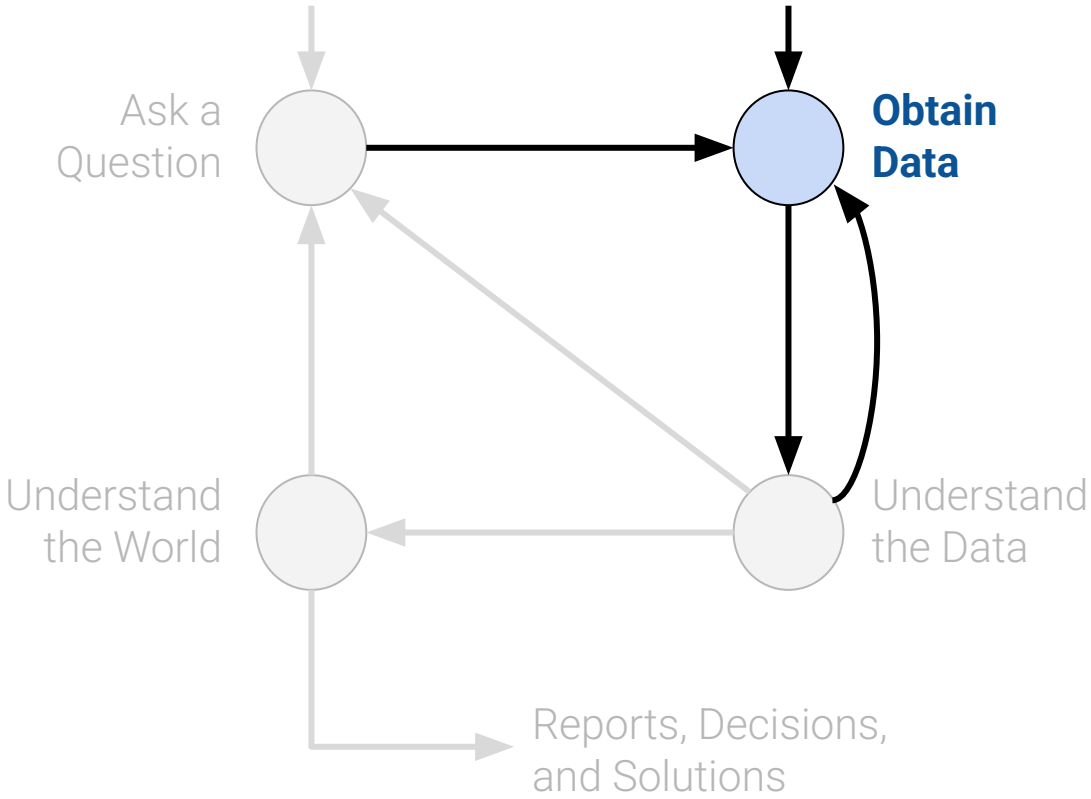
2. Data Acquisition and Cleaning

- What data do we have and what data do we need?
- How will we sample more data?
- Is our data representative of the population we want to study?



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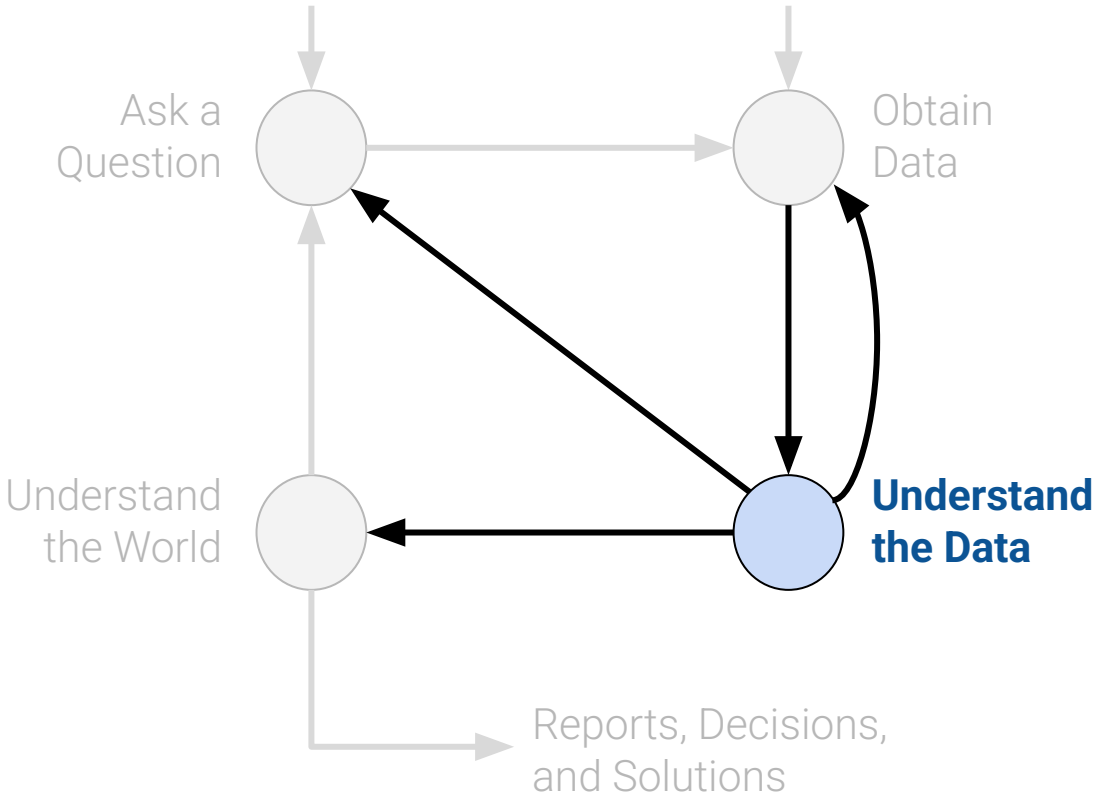
3. Exploratory Data Analysis & Visualization

- How is our data organized and what does it contain?
- Do we already have relevant data?
- What are the biases, anomalies, or other issues with the data?
- How do we transform the data to enable effective analysis?



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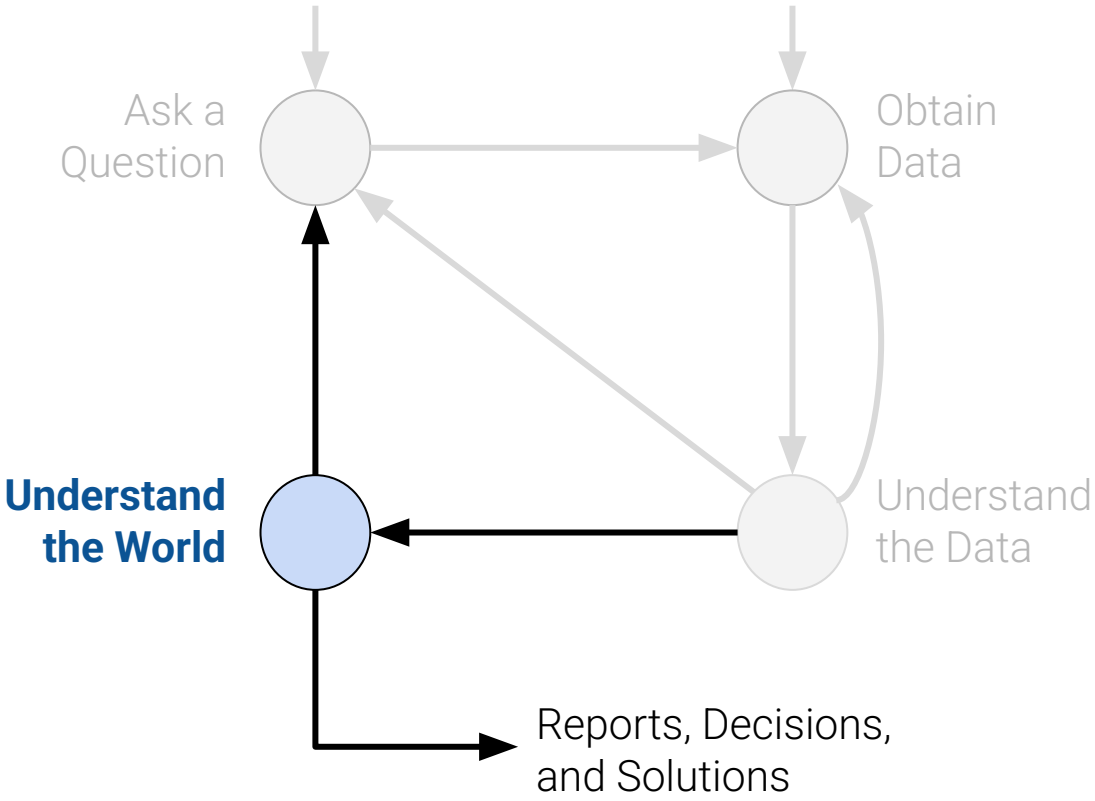
4. Prediction and Inference

- What does the data say about the world?
- Does it answer our questions or accurately solve the problem?
- How robust are our conclusions and can we trust the predictions?



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Participation Questions

- Questions will appear on these slides
- Answer questions on iClicker Quiz
  - "P01" for Participation Quiz 01.

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Q1

How many classes can you miss while still getting full participation marks?

A. 0

B. 2

C. 5

D. 7

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Q2

Which of the following is ***not*** a way to start the Data Science Lifecycle?

A. Ask a research question, then collect relevant data.

B. Obtain data, then find interesting patterns that suggest a research question.

C. Decide on a conclusion, then collect data that supports that conclusion.

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Q3

Which of the following is a potential indication that the Data Science Lifecycle has gone wrong?

A. Our exploration of the data gives us results we didn't expect.

B. A colleague points out a different way to clean our data, which results in very different results.

C. Our reports must include disclaimers about biases present in our data.

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Q3 Notes

- We should always be open to conclusions that we didn't expect!

- There are problems with almost all data sets.
  - Including a disclaimer is intellectually honest!
  - Reports *without* a discussion of potential biases are sus.

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Q4

Which of the following is a "good" question?

A. Should we abolish Daylight Savings Time (DST)?

B. Would people be better off if we abolished DST?

C. What's the relationship between DST and car crashes?

D. Would abolishing DST lead to a reduction in car crashes in the three business days after the time change?

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Q4 Notes

Research questions should:

- Be able to be answered with a well-collected data set.
  - E.g. *Not* "are people better off".
- Hint at exactly which data would be collected.
  - E.g. number of car crashes
- Have a definitive answer that suggests an action
  - E.g. "Yes, the data show an increase in car crashes."

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